

DFT - Part 1

- Introduction
- Internal Scan

- ◆ FF-based

- * MUXed-D scan (1973, Stanford)

- MUXed-D scan flip-flop
 - Test Mode Operation
 - Ckt. Model for ATPG

- ◇ SSF

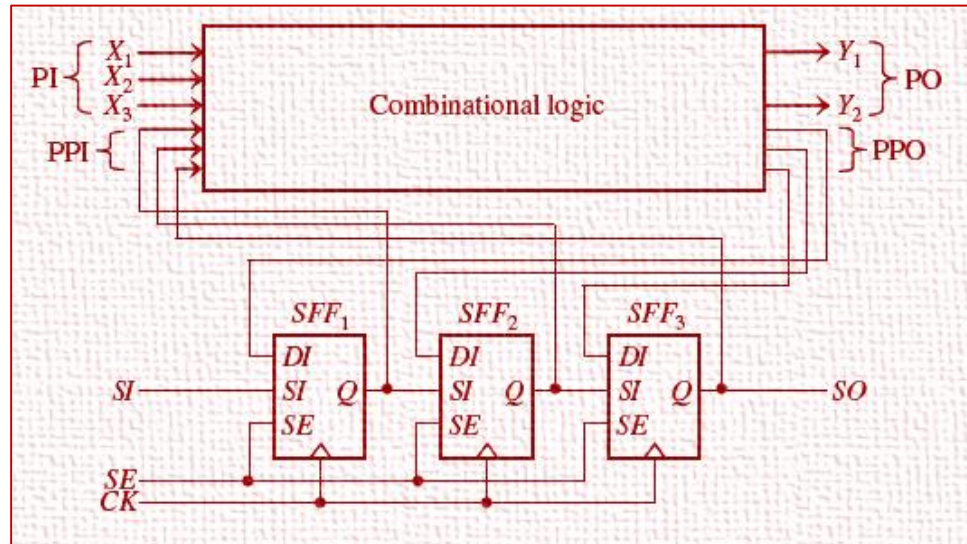
- ◇ LOS

- ◇ LOC

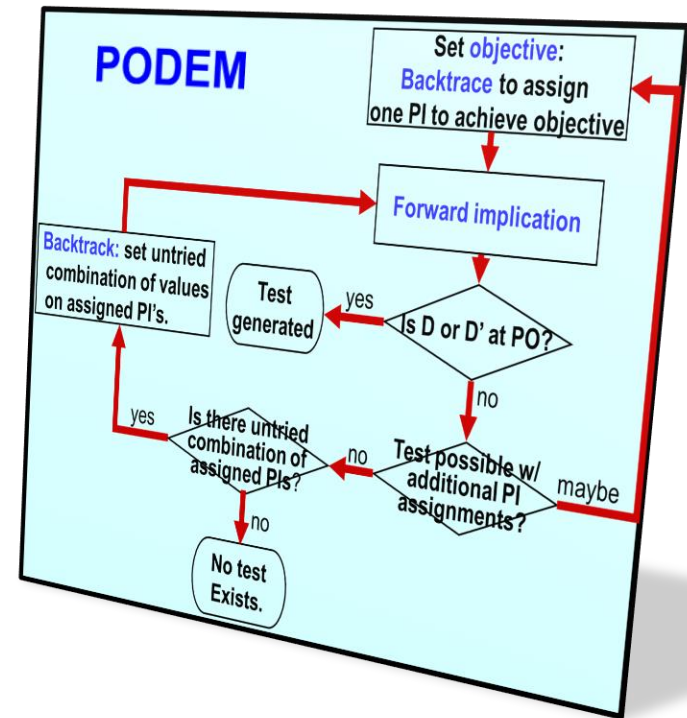
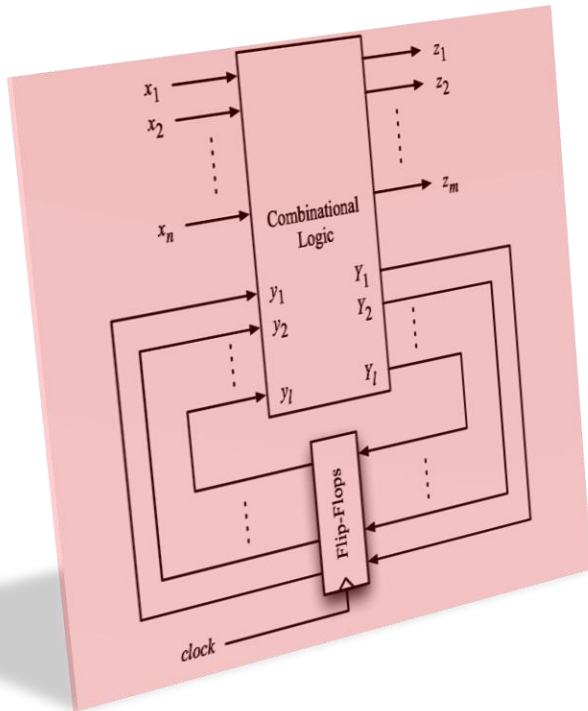
- * Clocked scan

- * Other scan

- ◆ Latch-based

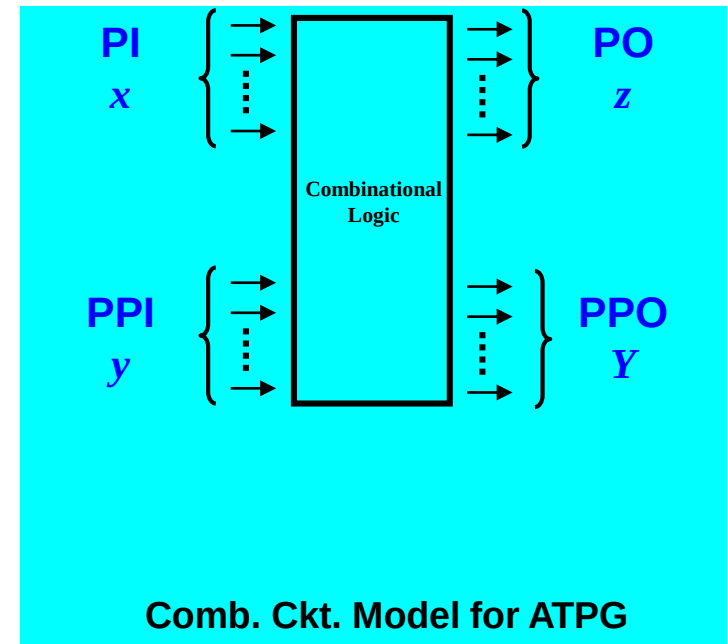
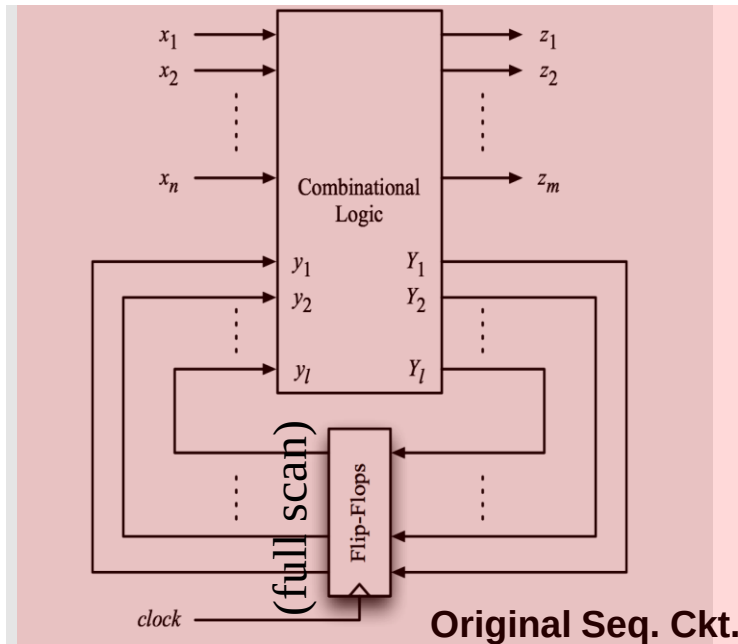


How to Run Comb. ATPG on Seq. Ckt?



DFT Turns Seq. Ckt. to Comb. Ckt.

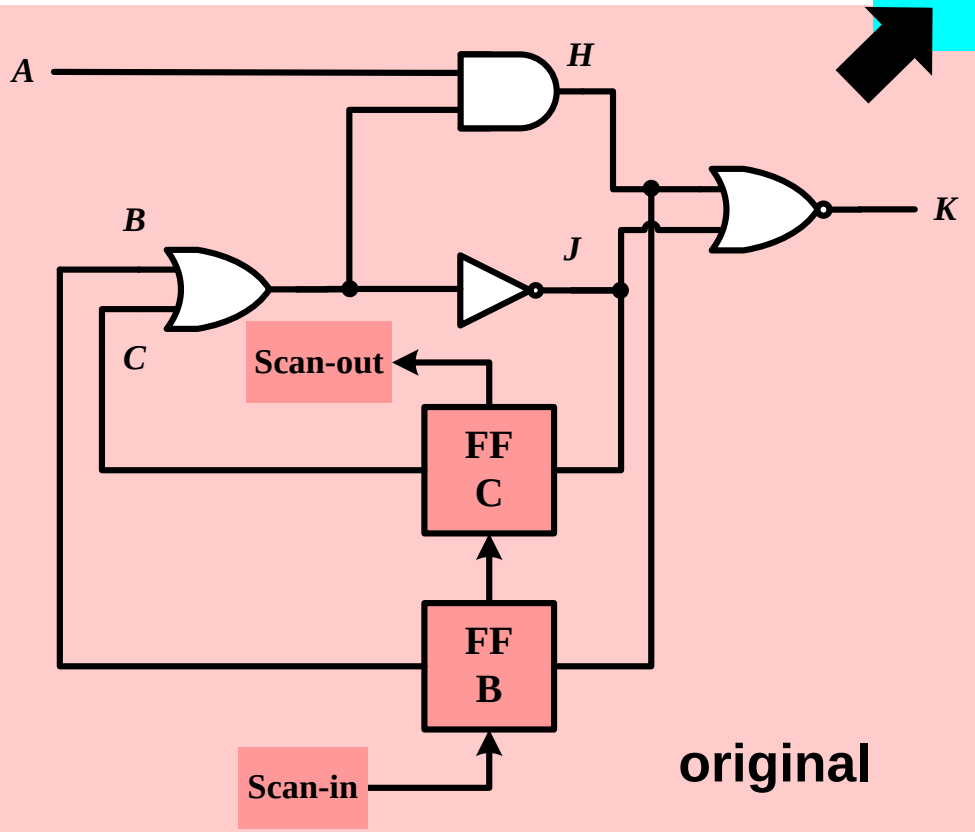
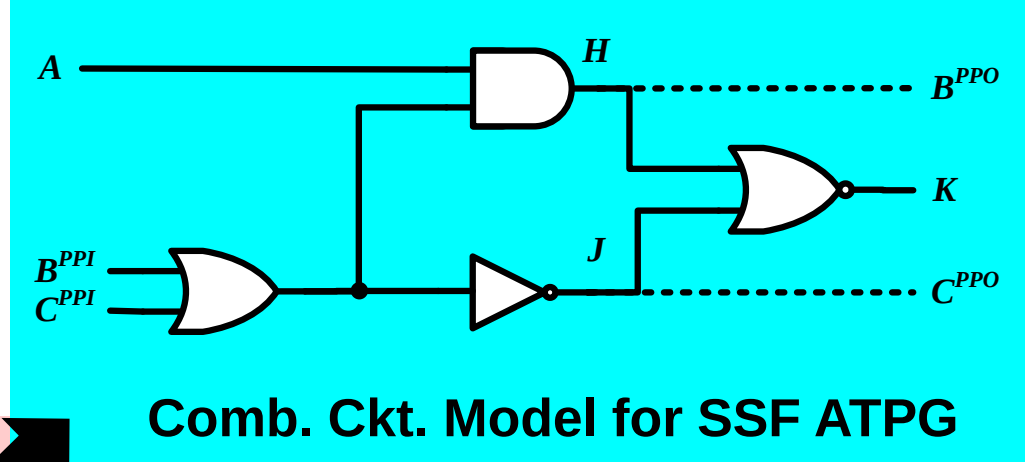
- DFT turns sequential ckt into combinational ckt in test mode
 - ♦ ATPG sees only **comb. ckt. model**
- Scan FF become **Pseudo Primary Input (PPI)**, fully controllable
- Scan FF become **Pseudo Primary Output (PPO)**, fully observable



Comb. ATPG Much Faster than Seq. ATPG

Example (1/2)

- Two scan FF in a scan chain
- SI → FF-B → FF-C → SO

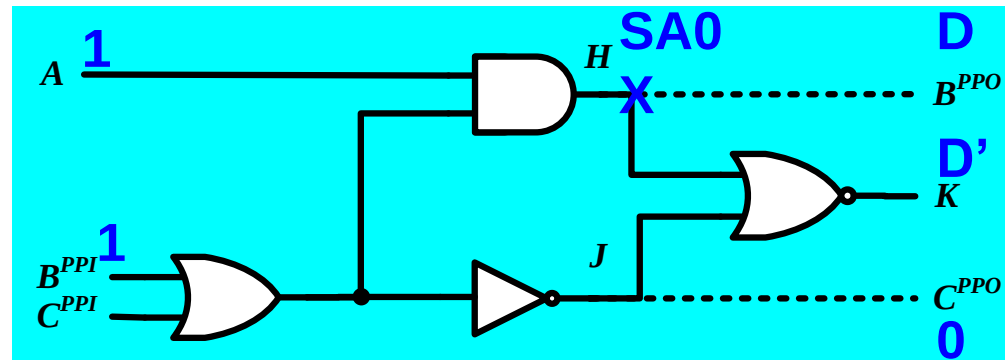


- A is **PI**
- B,C are **PPI**
- K is **PO**
- HJ are **PPO**
 - ◆ $H=B^{PPO}, J=C^{PPO}$

**NOTE: this model assume
no fault in FF. (see FFT)**

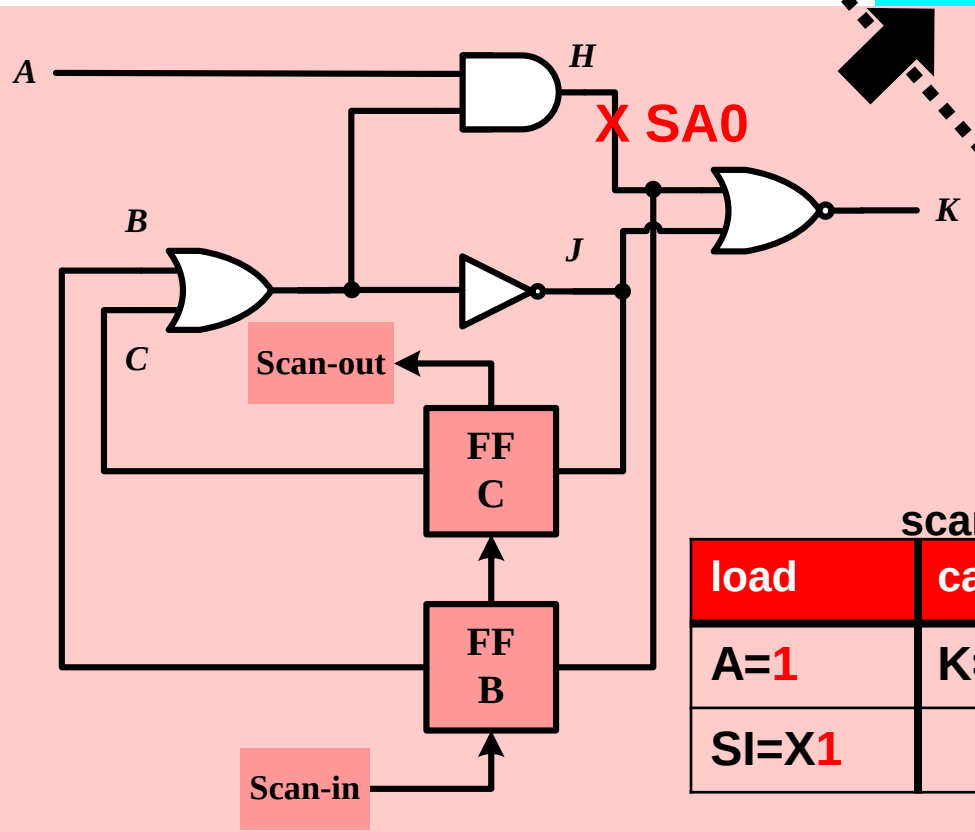
Example (2/2)

- Generate SSF ATPG pattern
 - ◆ H SA0 fault



PI	PPI	PO	PPO
A	$B^{PPI}C^{PPI}$	K	$B^{PPO}C^{PPO}$
1	1 X	D'	D 0

comb. pattern



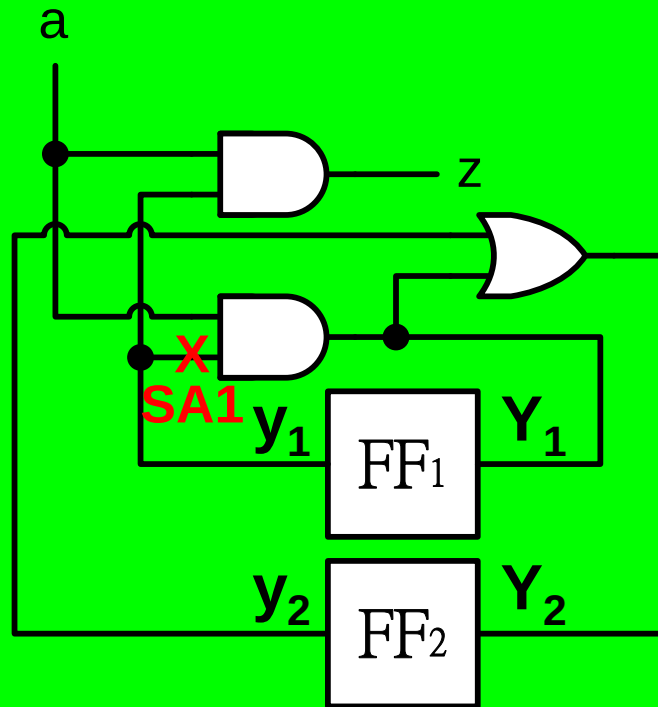
scan pattern

load	capture	unload
A=1	K=L	
SI=X1		SO=LH

Quiz

Q1: Convert seq. ckt. into comb. ckt. model for SSF ATPG.

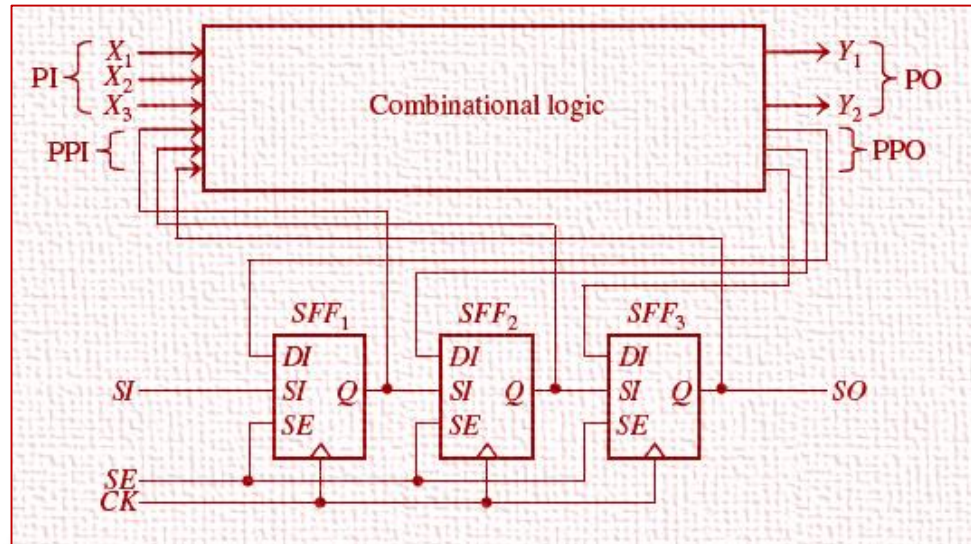
Q2: Generate a test pattern for SA1 fault.
(show comb. pattern)



PI	PPI	PO	PPO
a	y ₁ y ₂	z	Y ₁ Y ₂

DFT - Part 1

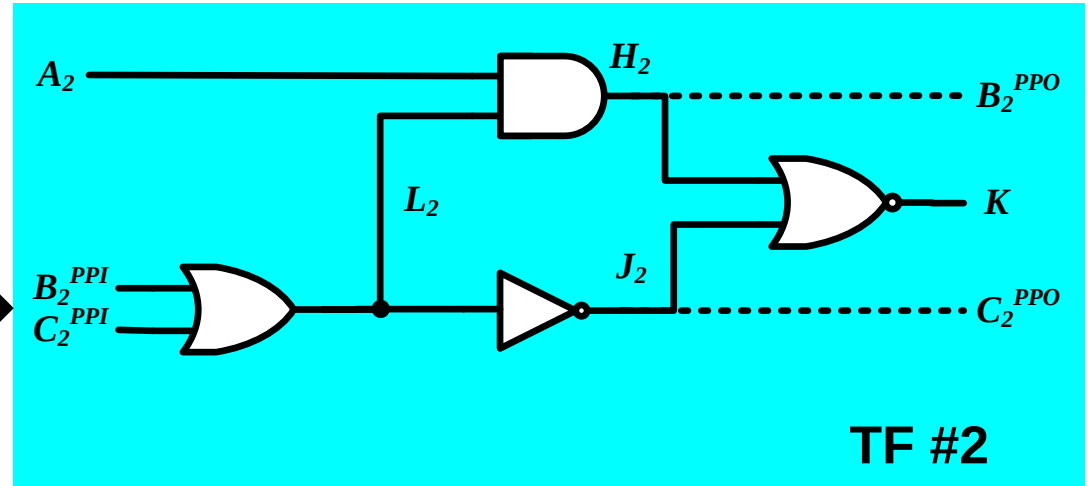
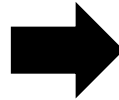
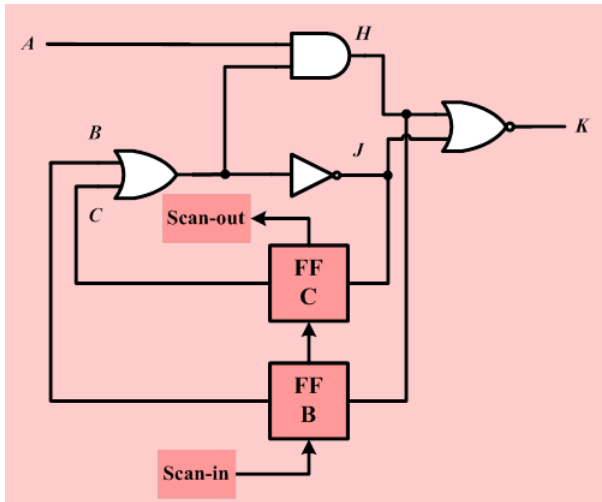
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 - Ckt Model for ATPG
 - ◇ SSF
 - ◇ LOS
 - ◇ LOC
 - * Clocked scan
 - * Other scan
 - ◆ Latch-based



Model for LOS ATPG

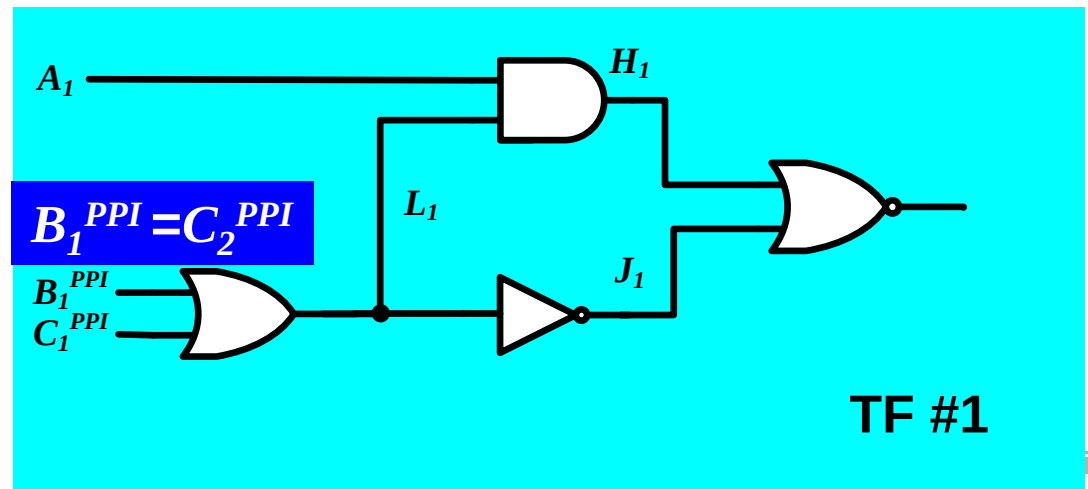
- LOS for Transition Delay Fault

- ♦ 1. Generate SSF pattern V_2 in time frame TF#2
- ♦ 2. Then apply constraints to generate V_1 in time frame TF#1



TF #2

Subscripts = time frame
Why TF#2 first? see FFT



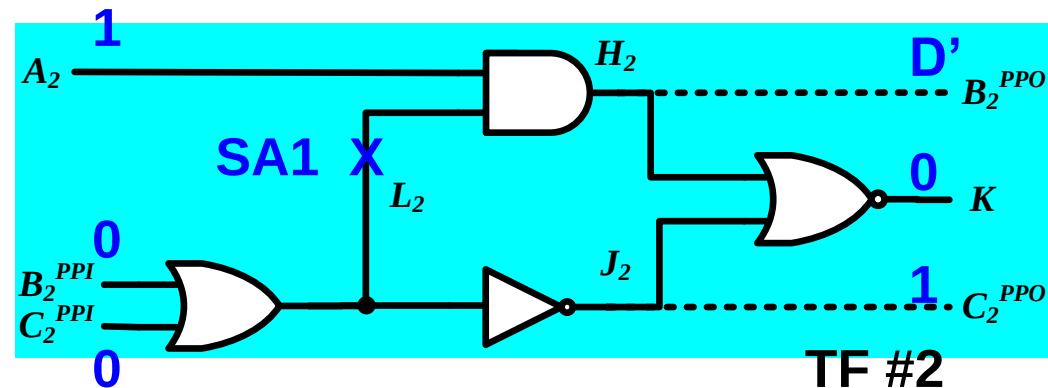
TF #1

LOS Example (1/2)

- Example: L slow-to-fall (STF) fault

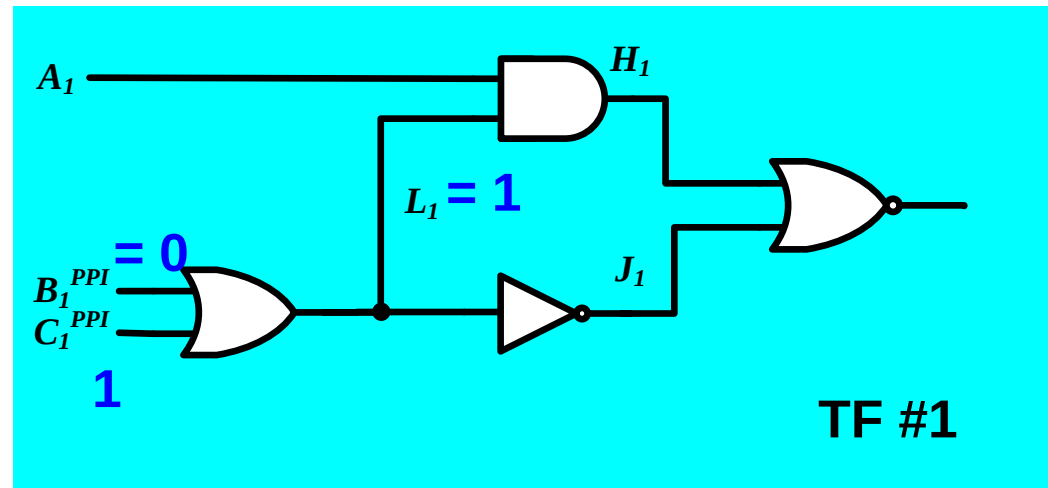
Time frame #2:

- Inject L_2 SA1
- $B_2^{PPI}=0$, $C_2^{PPI}=0$, $A_2^{PI}=1$
- $B_2^{PPO} = D'$



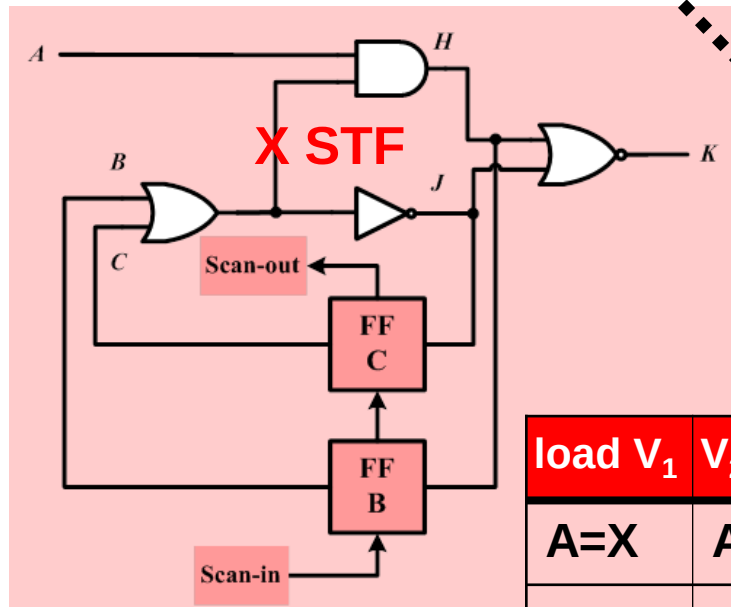
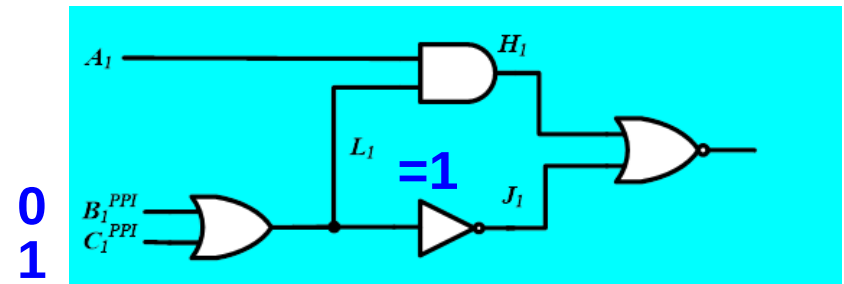
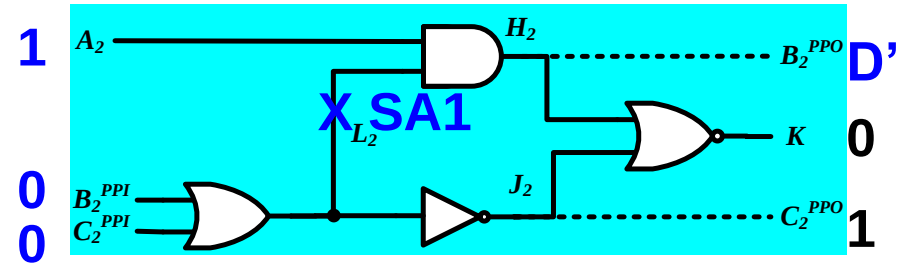
Time frame #1:

- STF fault constraint: $L_1 = 1$
- LoS constraint: $B_1^{PPI}=C_2^{PPI}=0$
- so, $C_1^{PPI}=1$



Only One Time Frame Memory Needed

LOS Example (2/2)



V_1		V_2		PO	PPO
A	$B_1^{PPI} C_1^{PPI}$	A	$B_2^{PPI} C_2^{PPI}$	K	$B_2^{PPO} C_2^{PPO}$
X	0 1	1	0 0	0	D' 1

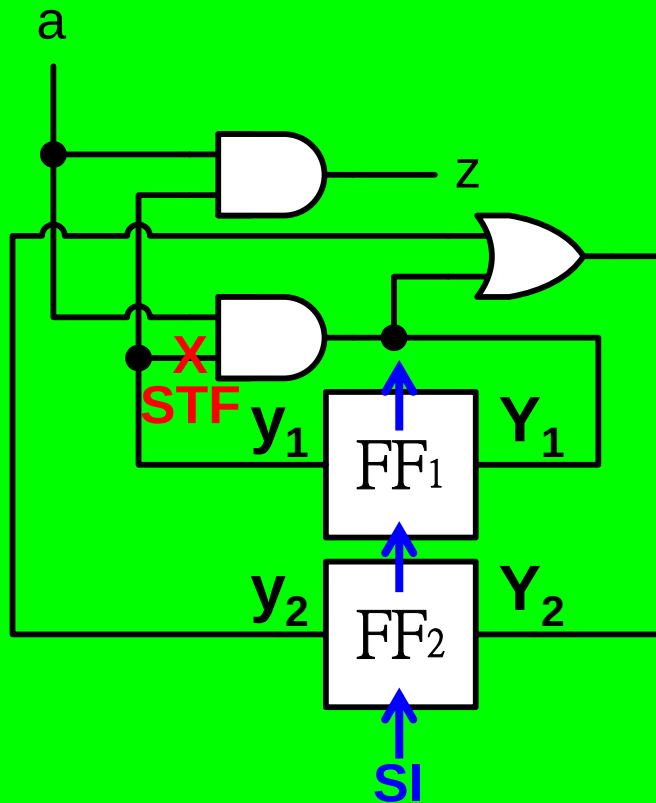
load	V_1	V_2	capture	unload
A=X	A=1	K=L		
SI=10	SI=0			SO=HL

Quiz

Q: Generate LOS test pattern for STF fault.

suppose $SI \rightarrow FF_2 \rightarrow FF_1 \rightarrow SO$

(show comb. pattern)



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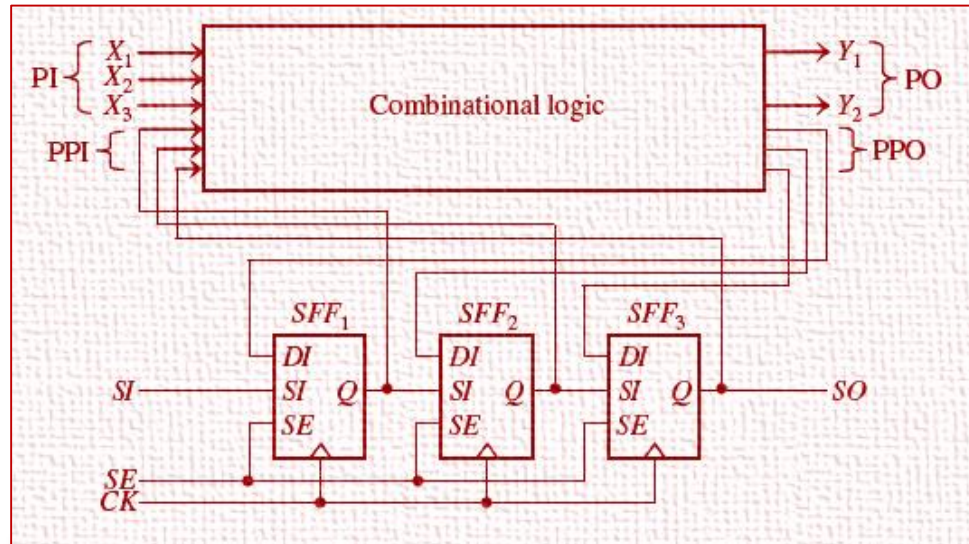
- ◇ LOS

- ◇ LOC

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- ◆ Latch-based

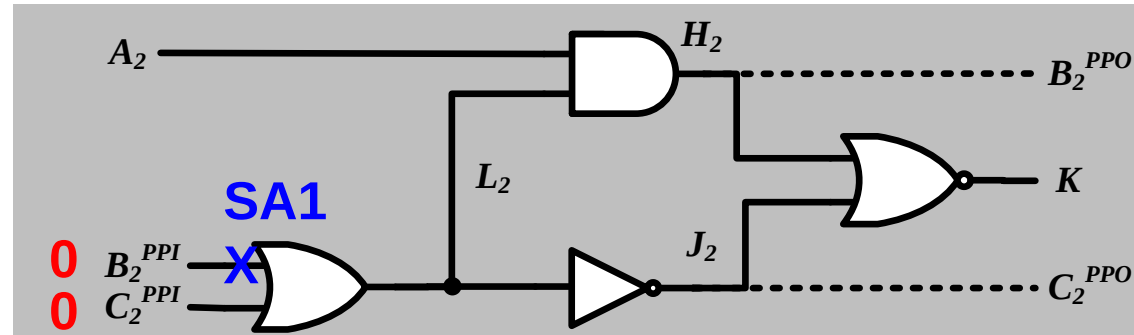


LOS Untestable

- Example: *B slow-to-fall (STF)* fault

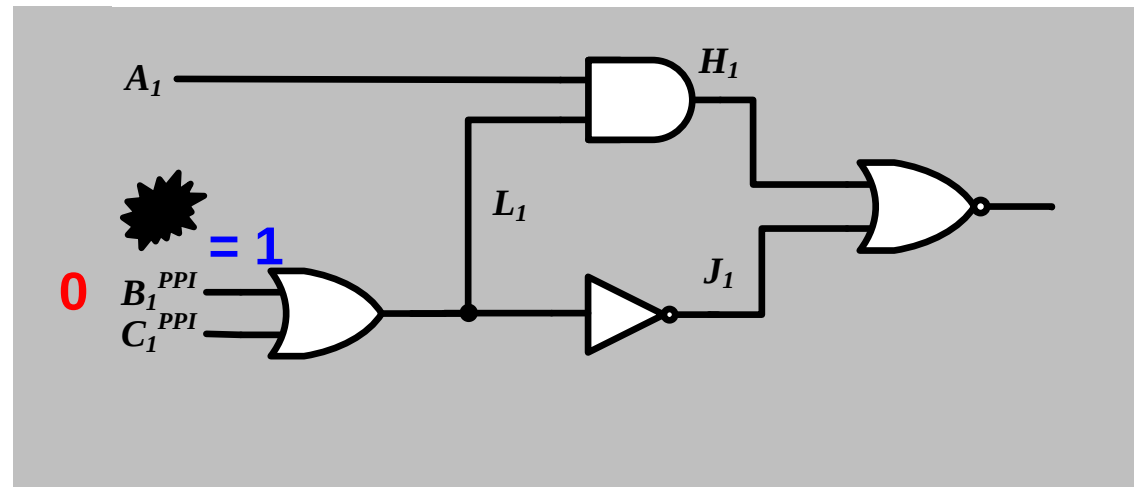
Time frame #2:

- Inject B_2^{PPI} SA1
- $B_2^{PPI}=0, C_2^{PPI}=0$
- $C_2^{PPO} = D$



Time frame #1:

- Fault constraint: $B_1^{PPI}=1$
- LoS constraint: $B_1^{PPI}=0$
- Conflict!

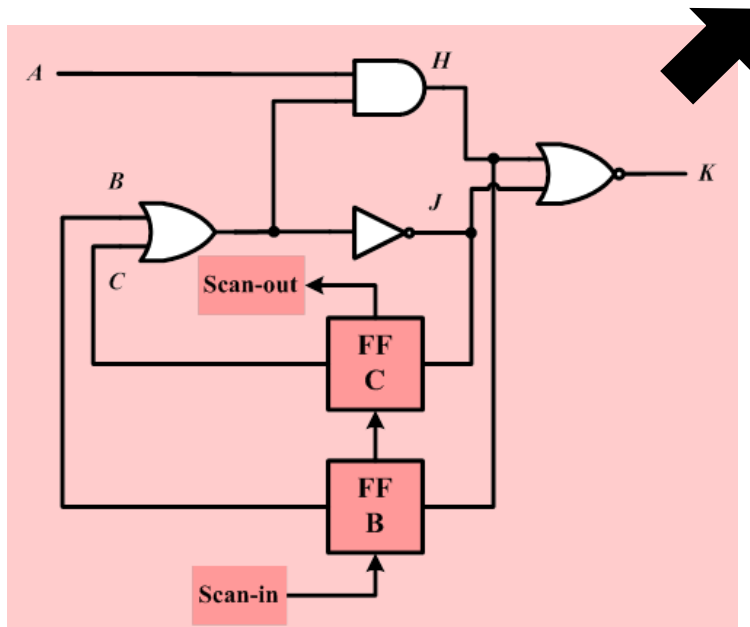
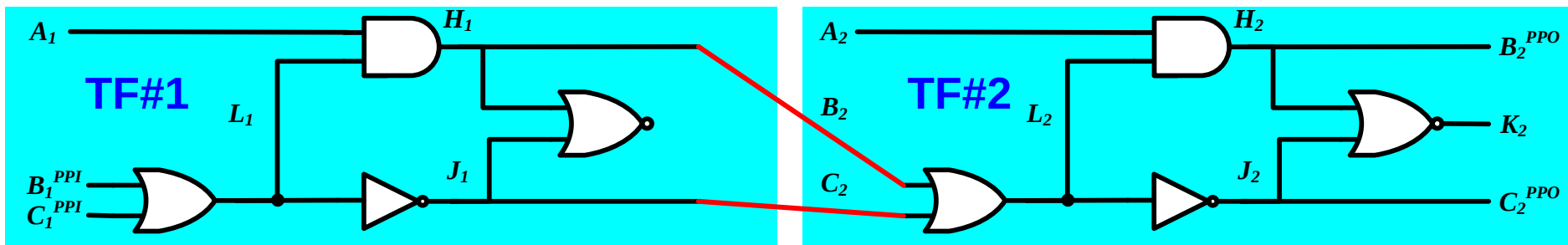


LOS Untestable due to Structural Dependency

Model for LOC ATPG

- LOC for TDF ATPG

- ♦ 1. Duplicate combinational Ckt into two copies: **TF#1** and **TF#2**
- ♦ 2. Connect TF#1 and TF#2 as one big ckt

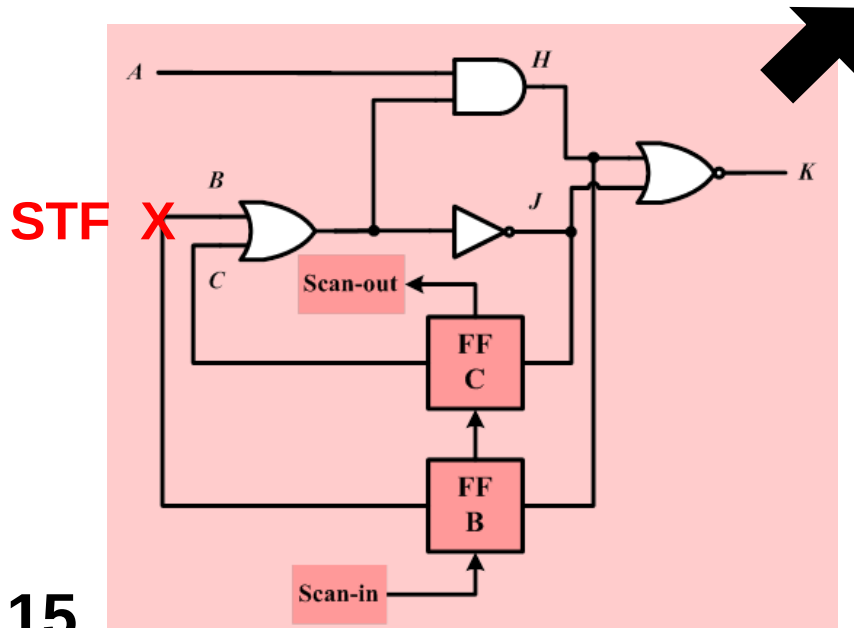
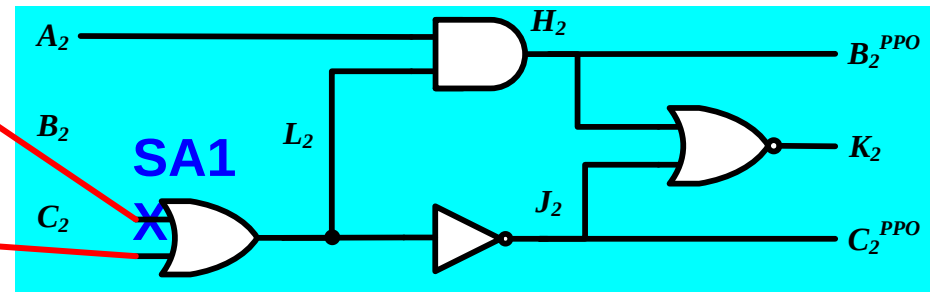
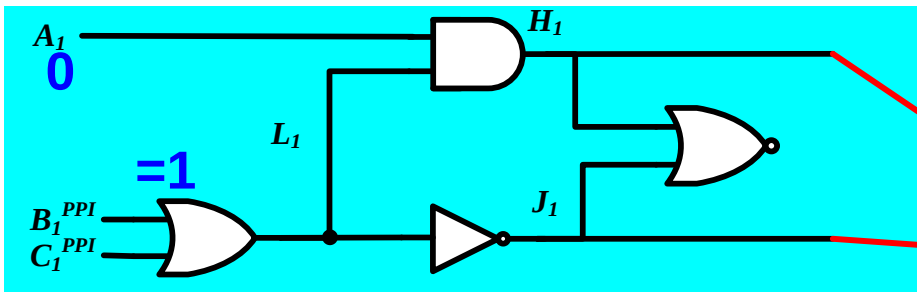


Like SeqATPG but with constraints. These constraints

**LOC Needs
Time Frame Expansion**

LOC Example (1/2)

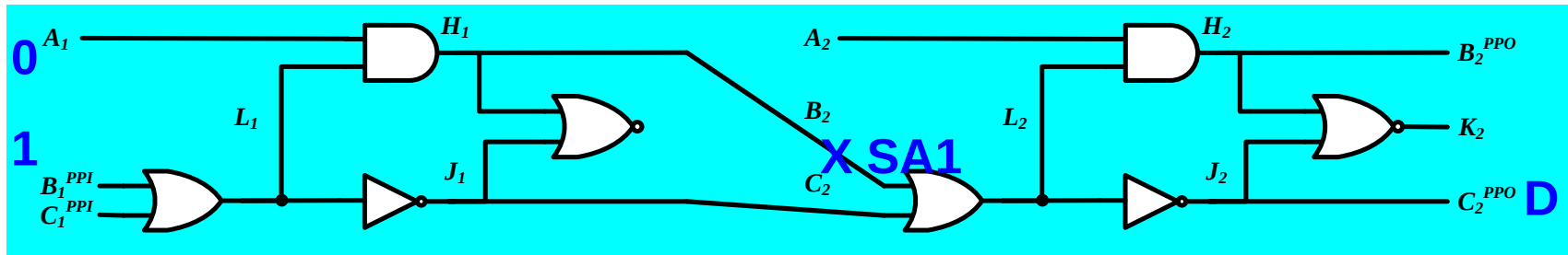
- Example: *B slow-to-fall (STF)* fault
 - ♦ SSF ATPG B_2 SA 1 fault
 - ♦ with constraint $B_1^{PPI}=1$



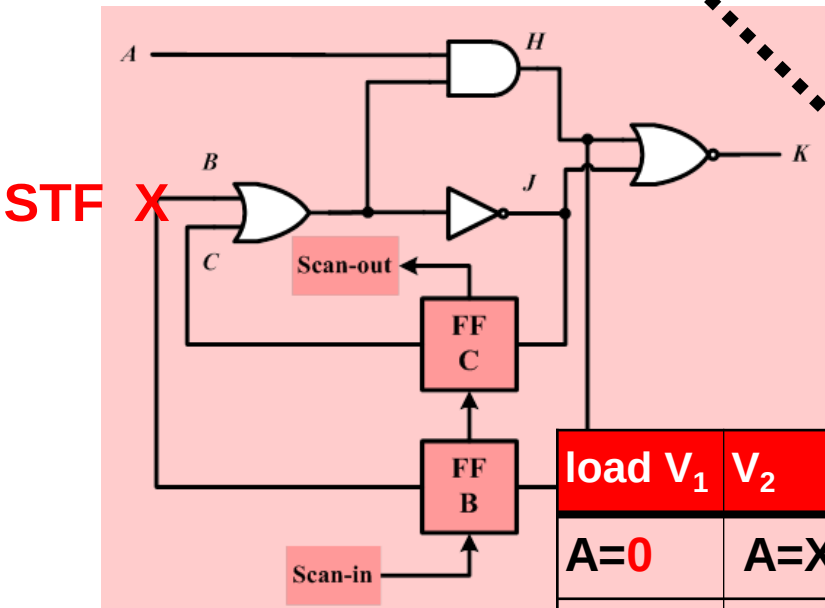
- PODEM
- Objective: $B_2=0$
 - ♦ Backtrace $A_1=0$
- Objective: $C_2=0$
 - ♦ Backtrace $B_1^{PPI}=1$
- Simulate $C_2^{PPO} = D$

Now It is LOC Testable!

LOC Example (2/2)



V ₁		V ₂	PO	PPO
A ₁	B ₁ ^{PPI} C ₁ ^{PPI}	A ₂	K	B ₂ ^{PPO} C ₂ ^{PPO}
0	1 X	X	X	X D

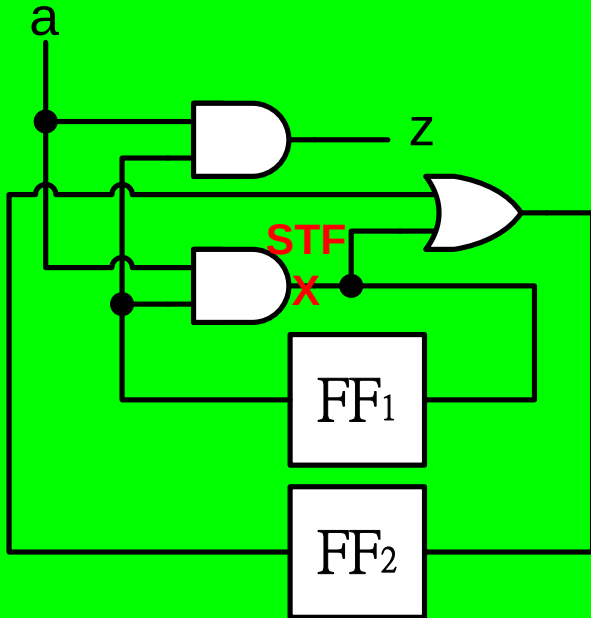


load V ₁	V ₂	capture	unload
A=0	A=X	K=X	
SI=X1			SO=HX

Quiz

Q1: Please draw ckt model for LOC

Q2: Generate LOC test pattern for STF fault.

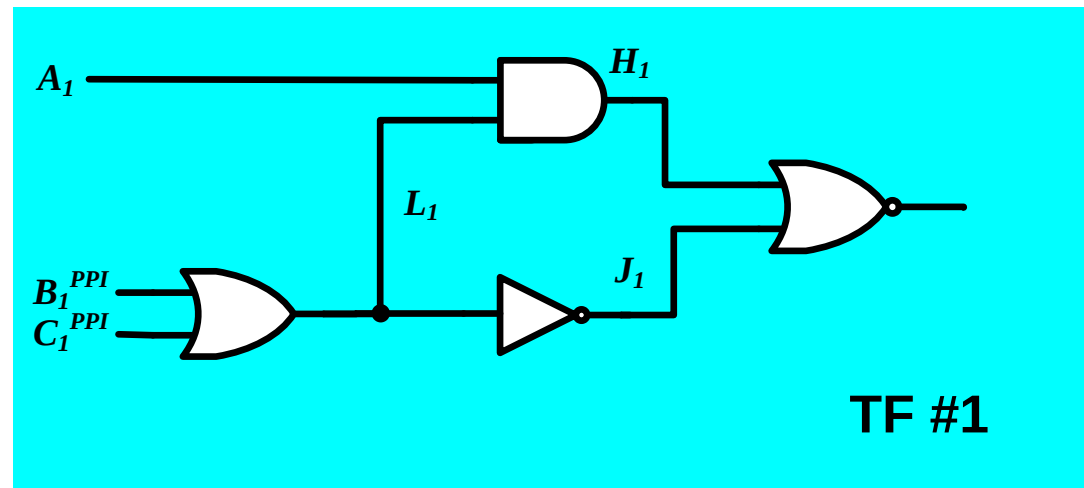
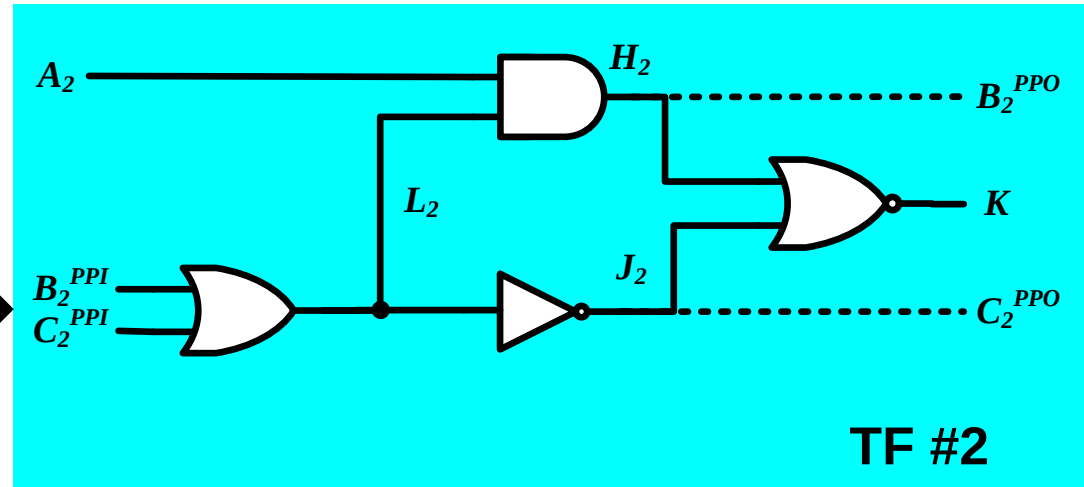
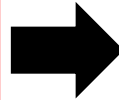
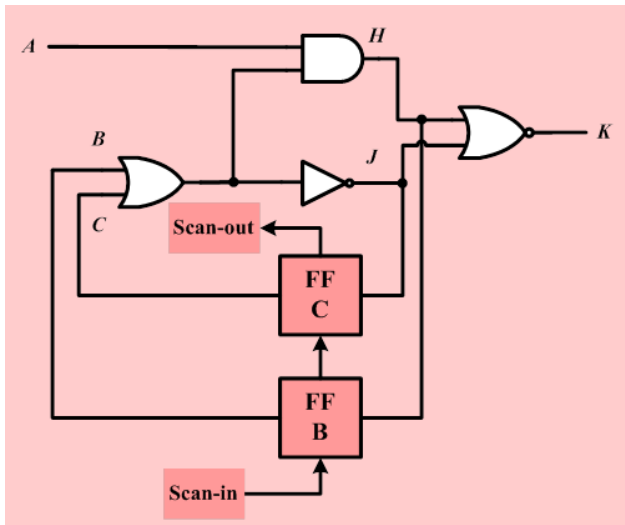


Summary

- Run comb. ATPG on seq. ckt with DFT
- SSF: All FF becomes PPI/PPO
- LOS
 - ♦ Generate V_2 first, then add constraints to V_1
 - 😊 Small memory. Fast run time
 - 😞 **Fault coverage limited** due to structural dependency problem
- LOC
 - ♦ Duplicate Ckt into two copies
 - 😊 Good FC
 - 😞 **More memory. Slow run time**
- Current practice: mix LOS and LOC
 - ♦ Do **LOS first**, save memory/computation time
 - ♦ **Then do LOC**, detect remaining faults

FFT

- Q: In LOS ATPG, why generate V_2 first? Why not V_1 first?
 - ♦ 1. Generate SSF pattern for V_2 ,
 - ♦ 2. Apply constraints to generate V_1



FFT 2

**Q: This model assumes FF are good.
What if FF are faulty?**

